Class Test: Containerize a Python Application (Containerization Focus)

Objective

The goal of this hands-on is to evaluate your understanding of **containerization concepts**:

- Base images
- Dependency installation
- Image naming and tagging
- Container execution and testing

This does NOT test Python, NumPy, or pip knowledge.

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Problem Statement

You are given a simple Python program that:

1. Takes a **SAP ID** as input
2. Compares it with a **stored SAP ID**
3. Prints:
   - `Matched` if both SAP IDs are same
   - `Not Matched` otherwise

You must **containerize this application** using Docker.

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Application Code (Provided)

Create a file named `app.py` with the following content:

```
# Simple program to verify SAP ID
# This program is intentionally basic
# Focus of this task is containerization, not Python
```

```
import numpy as np  # dependency for learning purpose

stored_sapid = "REPLACE_WITH_YOUR_SAPID"
user_sapid = input("Enter your SAP ID: ")

if user_sapid == stored_sapid:
    print("Matched")
else:
    print("Not Matched")
```

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Containerization Requirements (Strict)

1. Base Image

- Use an **official Python image**
- Python version: **3.10 or above**

2. Dependencies

- The program uses:
    - `python`
    - `pip`
    - `numpy`

These **must be installed inside the container**, not on the host system.

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3. Image Name and Tag

- Image name **must be**: `sapid-checker`
- Image tag **must be your SAP ID**

Example:

```
sapid-checker:500123456
```

4. Dockerfile Requirements

Your Dockerfile must:

- Set a working directory
- Copy the application file
- Install required dependencies
- Run the Python program when the container starts

Hints (Read Carefully)

- Python images already include `pip`
- NumPy must be installed using `pip`
- Use `CMD` or `ENTRYPOINT` correctly
- The container should accept input from the terminal
- Use interactive mode while running the container

Build Instructions (Expected)

```
docker build -t sapid-checker:<YOUR_SAPID> .
```


Run & Test Instructions (Expected)

```
docker run -it sapid-checker:<YOUR_SAPID>
```

Example Output

```
Enter your SAP ID: 500123456
Matched
```

or

```
Enter your SAP ID: 500999999
Not Matched
```

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Evaluation Criteria

Criteria	Marks
Correct Dockerfile	✓
Correct base image	✓
NumPy installed inside container	✓
Correct image name & tag	✓
Container runs successfully	✓

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What is NOT evaluated

- Python syntax depth
- NumPy usage complexity
- Input validation logic

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Bonus (Optional – Not Mandatory)

- Use `.dockerignore`
- Reduce image size
- Explain each Dockerfile instruction in comments